Education

Institut Polytechnique de Paris (ENSAE Paris), Statistics — Palaiseau, France
2025 – 2028

- Research topics
- Reinforcement Learning, Control.
- Advisors
- Vianney Perchet, Hugo Richard, Marc Abeille.

Experience

Criteo, Research Scientist — Paris, France
2024 – present
2 years

Working on Reinforcement Learning applied to Real Time Bidding. Developing bidding algorithms that take into account exogenous information/high-dimensional contexts.

Criteo, Research Intern — Paris, France
May 2024 – Oct 2024
7 months

Worked on Local Differential Privacy applied to advertising, led to a publication to AISTATS.

Publications

Distribution-aware mean estimation under user-level local differential privacy.
We study an online resource allocation problem in which at each round, a budget B must be allocated across K arms under censored feedback. An arm yields a reward if and only if two conditions are satisfied (i) the arm is activated according to an arm-specific Bernoulli random variable with unknown parameter, and (ii) the allocated budget exceeds a random threshold drawn from a parametric distribution with unknown parameter. Over T rounds, the learner must jointly estimate the unknown parameters and allocate the budget so as to maximize cumulative reward facing the exploration–exploitation trade-off. We prove a worst-case regret lower bound Ω(T¯¯¯√) for the general model, even with known uniform threshold distributions. We then propose RA-UCB, an optimistic algorithm that leverages non-trivial parameter estimation and confidence bounds. When the budget B is known at the beginning of each round, RA-UCB achieves a regret bound of order ⌧̃(T¯¯¯√), and even ⌧(polylogT) under stronger assumptions. As for unknown, round dependent budget, we introduce \texttt{MG-UCB}, which allows within-round switching and infinitesimal allocations, and matches the regret guarantees of RA-UCB. We then validate our theoretical results through experiments on real-world datasets.
Corentin Pla, Hugo Richard, and Maxime Vono
arxiv.org/abs/2602.06565

Skills

Programming

Languages

French
Native speaker

English
Fluent